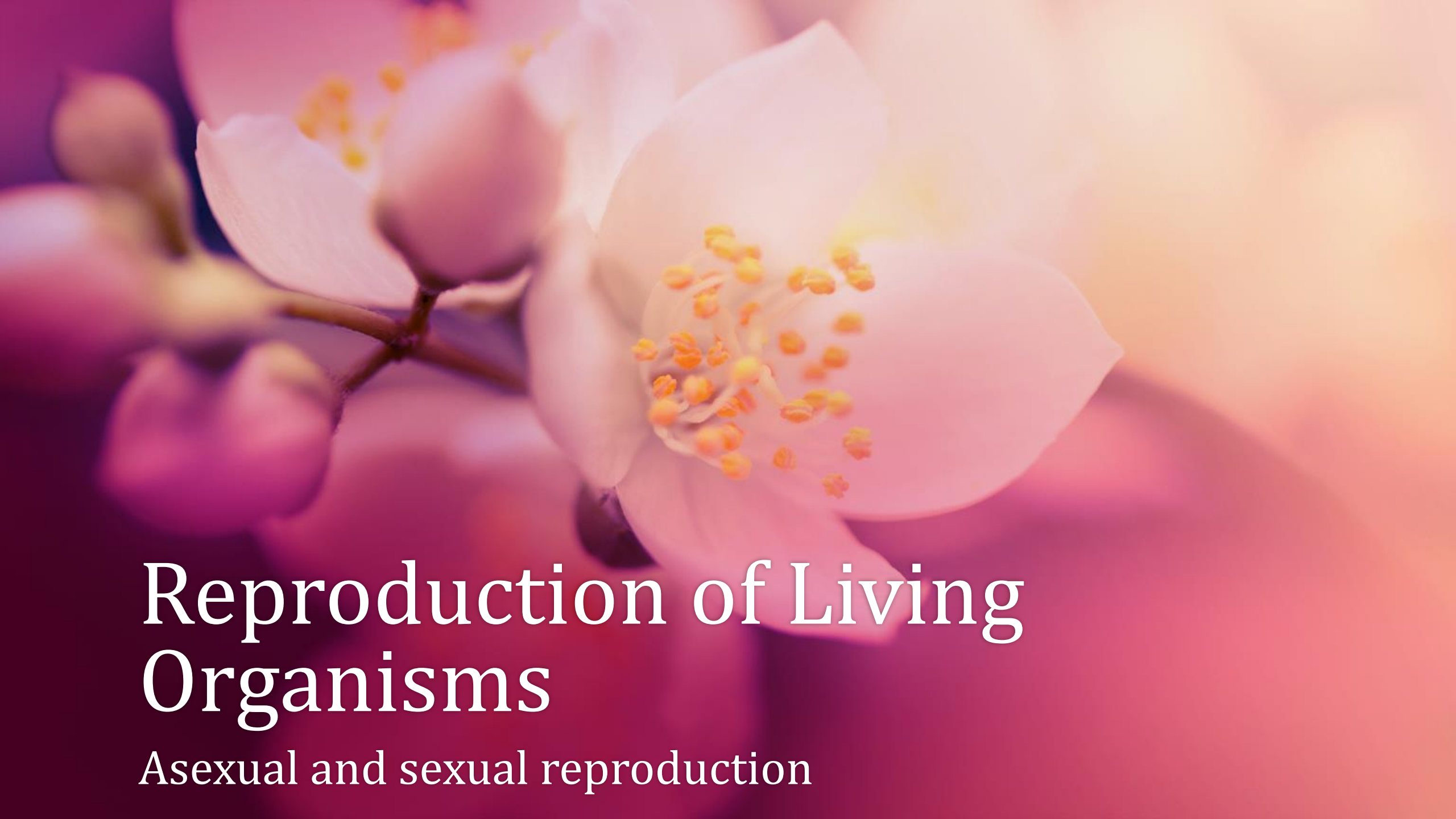




Reproduction of Living Organisms

Asexual and sexual reproduction

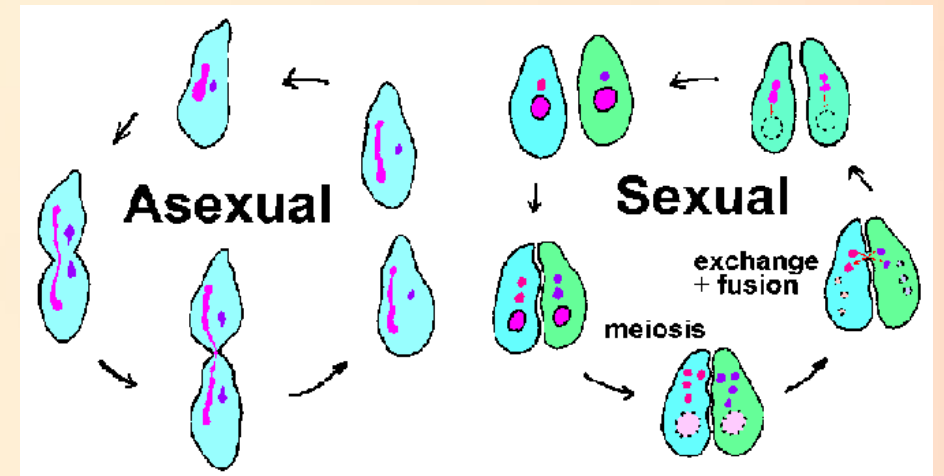
The Purpose of Reproduction

Purpose: Ensures the survival of a species.

- **Definition:** Reproduction is the process by which an organism produces offspring of the same kind.

There are **2 types of reproduction**:

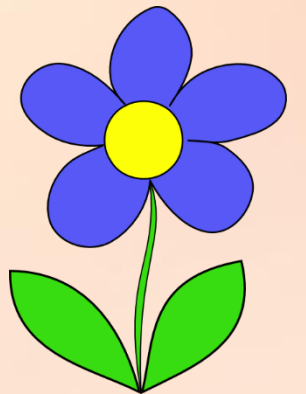
- Asexual
- Sexual



Some species reproduce by asexual reproduction and others by sexual reproduction

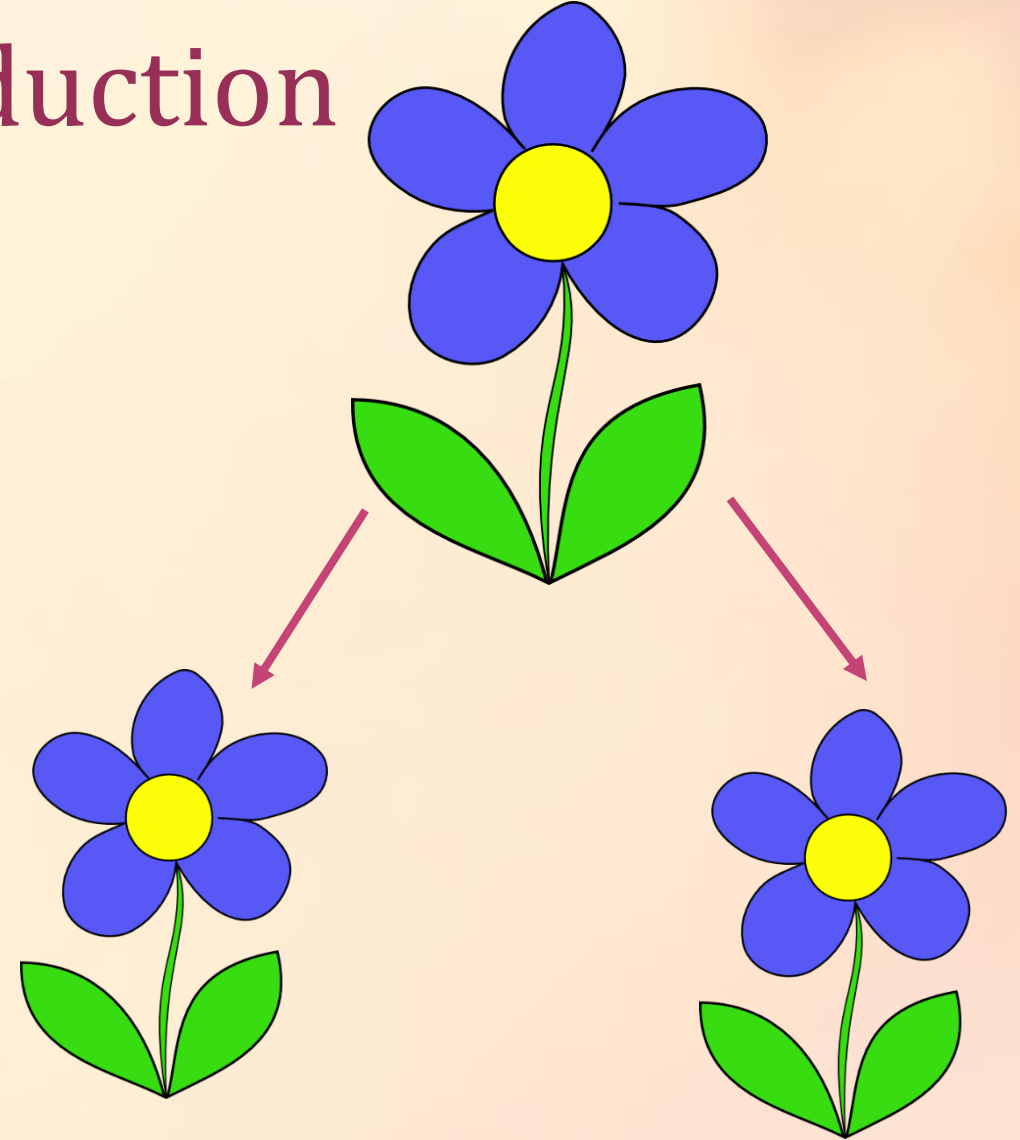
Asexual Reproduction

- In asexual reproduction, a new organism (sometimes more than one), is produced **from only one organism**. In other words, only one living organism is involved in the reproduction process.
- It is a form of reproduction which **does not** involve fertilization. In other words, it does not depend on male and female parts.
- Produces offspring that are identical to their parent (have the same physical and behavioural characteristics).
- Asexual reproduction = one parent.



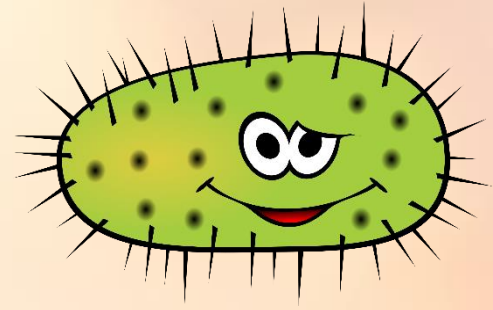
Summary of Asexual Reproduction

- One parent
- Offspring are identical to their parent.



Who reproduces asexually?

- Asexual reproduction is the primary form of reproduction for **single-celled organisms** (ex: **bacteria**).



- Many **plants and fungi** reproduce asexually as well.



- Some animals reproduce asexually as well (**sponges, hydras**).



Advantages and disadvantages of asexual reproduction

Advantages:

- Fast
- Easy

Disadvantage:

- Same DNA (genetically identical to their parent), therefore, there is no variation!

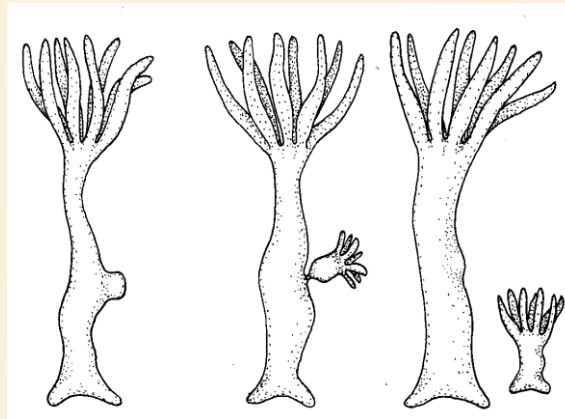
Types of Asexual Reproduction

There are 3 types of asexual reproduction:

- Budding
- Binary Fission
- Regeneration

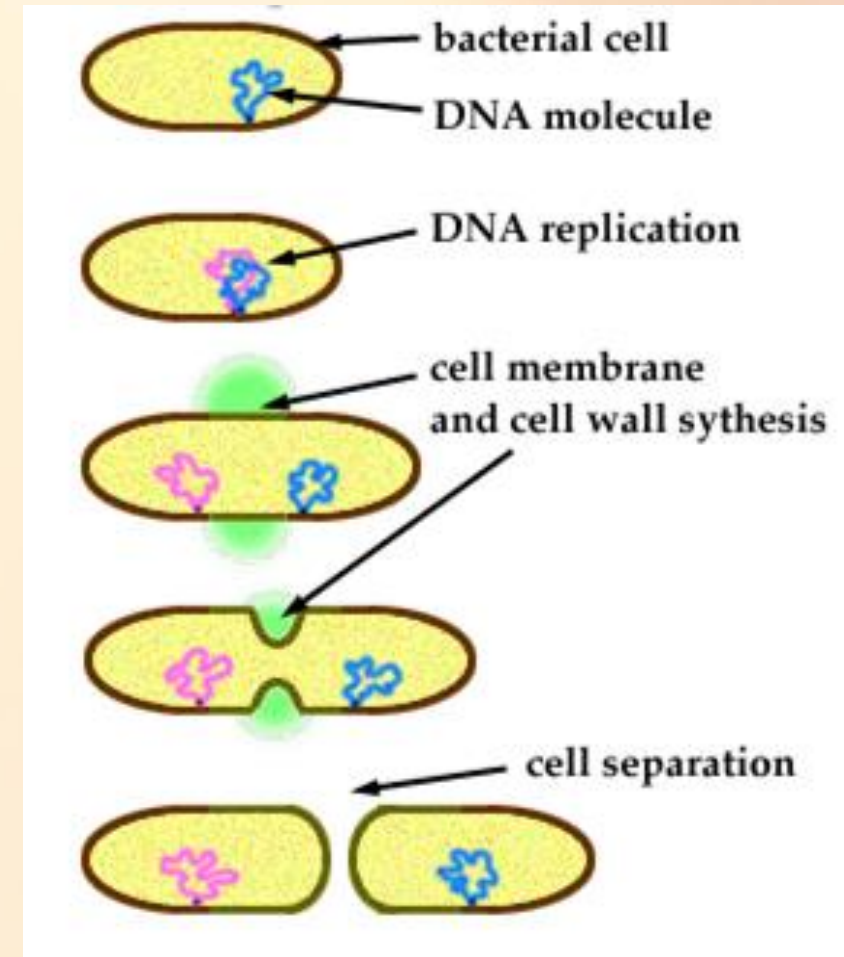
1. Budding

- Process by which a new-duplicate plant or animal begins to form at the side of the parent and enlarges until an individual is created (separates).
- Very common in plants.



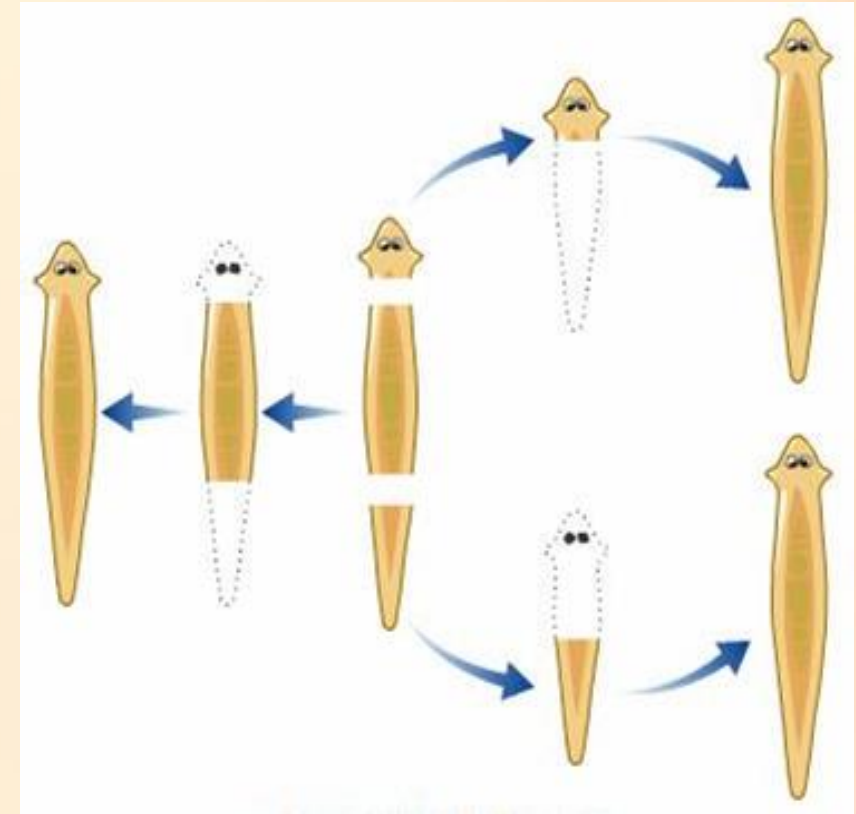
2. Binary Fission

- Becoming two by division of the complete organism.
- A type of cell division.
 - DNA replicates
 - Cell grows and splits
 - End result: 2 identical cells
- This happens with **bacteria & viruses**.
- This process is also used to create new cells in humans & animals.



3. Regeneration (Fragmentation)

- Also known as fragmentation.
- The ability to restore lost or damaged tissues, organs or limbs.
- It is a common feature in invertebrates, like worms, starfish and some reptiles.



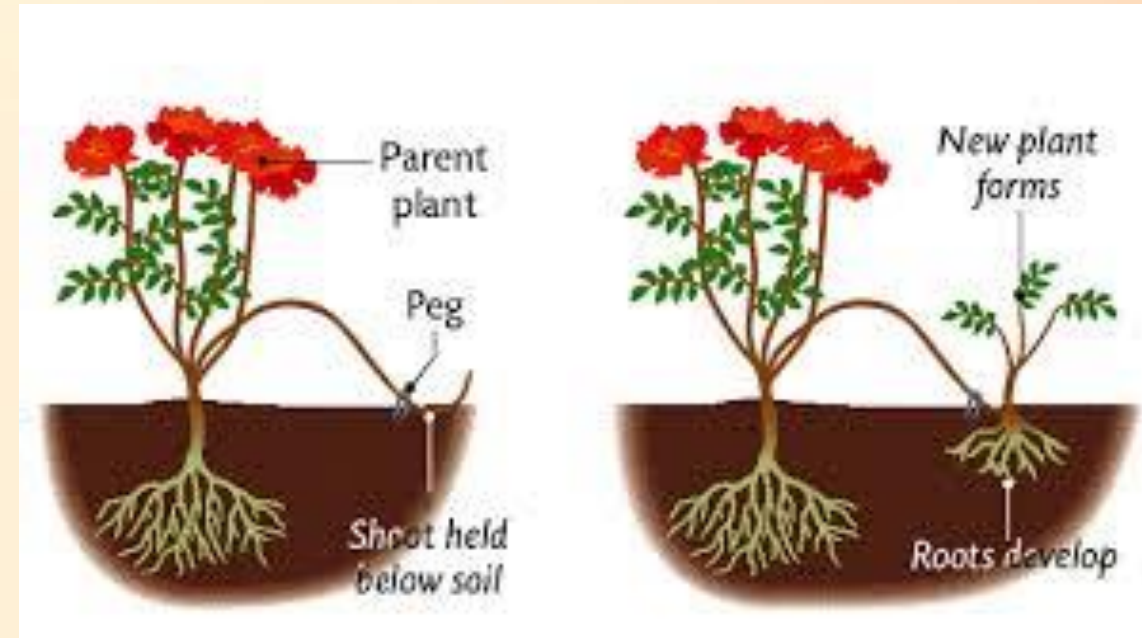
Artificial propagation of plants

- **Cutting:** This is a form of asexual reproduction in plants.
 - Using a knife, a small part of the parent plant (that has buds) is cut.
 - It is then buried in the soil.
 - After a few days the “cutting” (where you cut the plant) will start developing roots, and will grow into a new plant.



Artificial propagation of plants *continued...*

- **Layering**: this is a form of asexual reproduction in plants.
- A branch of the plant is pulled towards the ground and part of the branch is covered with soil leaving the end of the branch above the ground.
- With time, roots will develop out of the part of the branch that had been buried in the soil.
- Once the roots have been made, that part of the branch is cut and planted elsewhere so the new plant can grow.

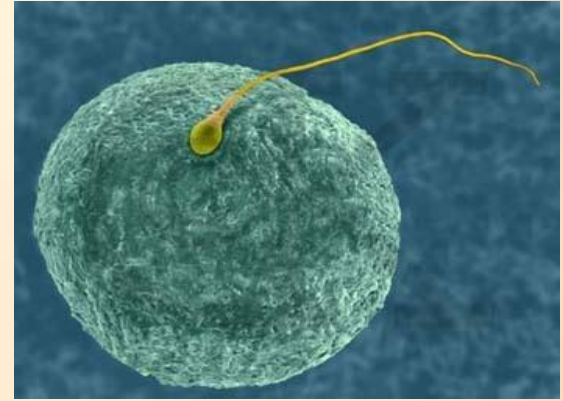


Sexual Reproduction

- Sexual reproduction requires the involvement of a male and a female parent.
- The offspring share many similarities with their parents, yet they each have a unique genetic makeup that is a combination of genes from both parents.



Sexual Reproduction



- Requires BOTH male and female gametes.
 - Gametes = sex cells
 - Sperm = male gamete
 - Ovum-egg = female gamete
- The ovum and sperm join to form an entirely new organism that is genetically different from both parents.
 - This results in increasing genetic diversity of the offspring.

Who Reproduces Sexually?

- Primary method of reproduction for the vast majority of living organisms, including:
 - almost all animals
 - flowering plants



Advantages & Disadvantages of sexual reproduction

- Advantages:
 - plants don't have to move.
 - mixes DNA.
- Disadvantage:
need external source for pollination to take place (wind, water, bee, bat, butterfly etc.).

Reproduction in plants

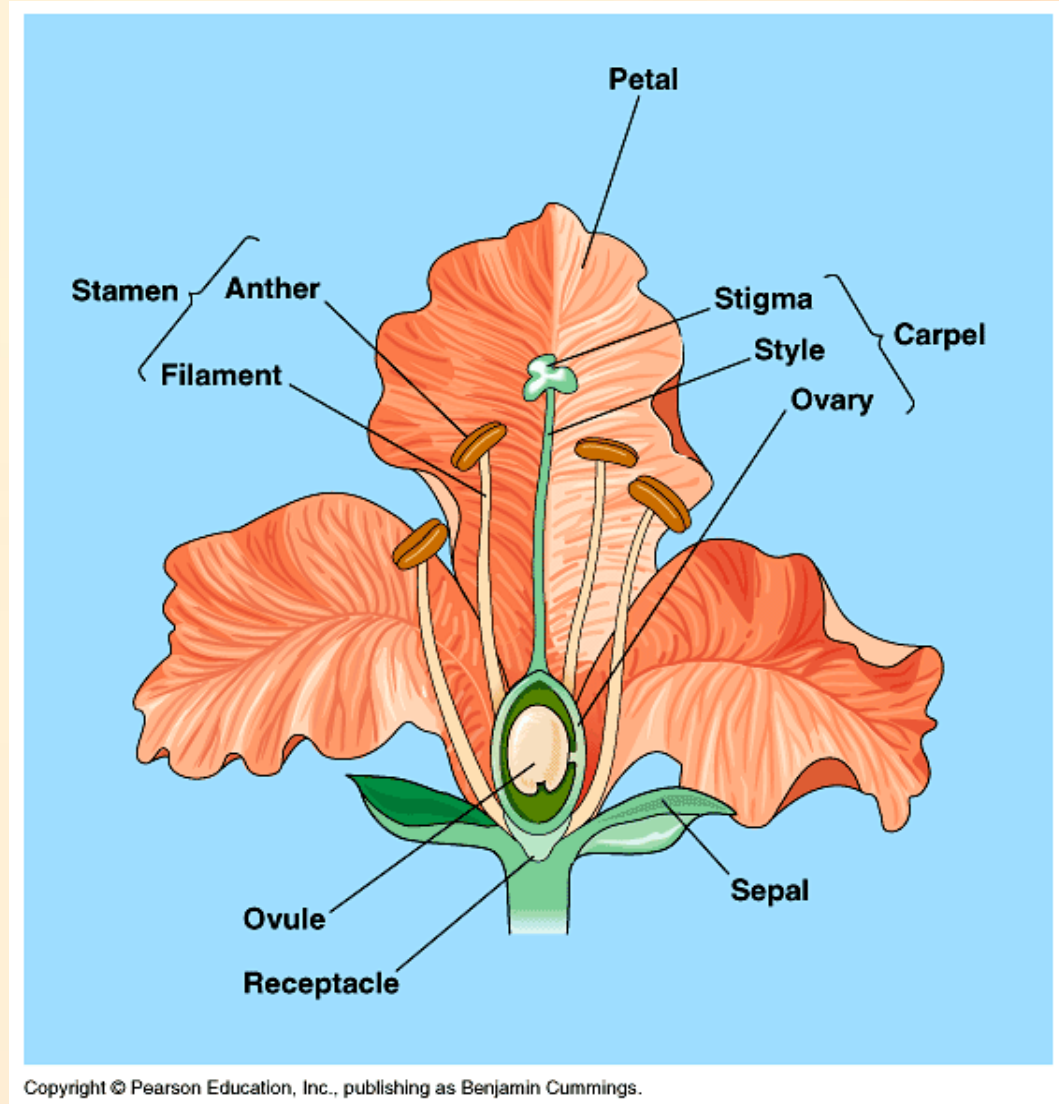
- Depending on the species, plants can reproduce both sexually and asexually.
- Example: the lilac tree



- The growth of several shoots at the base of the tree is an example of asexual reproduction.
- In the spring, this tree also grows many **flowers** that will produce seeds, which is an example of **sexual reproduction** in plants known as **flowering**.
- Therefore, this plant uses both asexual and sexual reproduction.

Anatomy of a Flower

1. Sepals
2. Petals
3. Stamens
4. Carpel (Pistil)

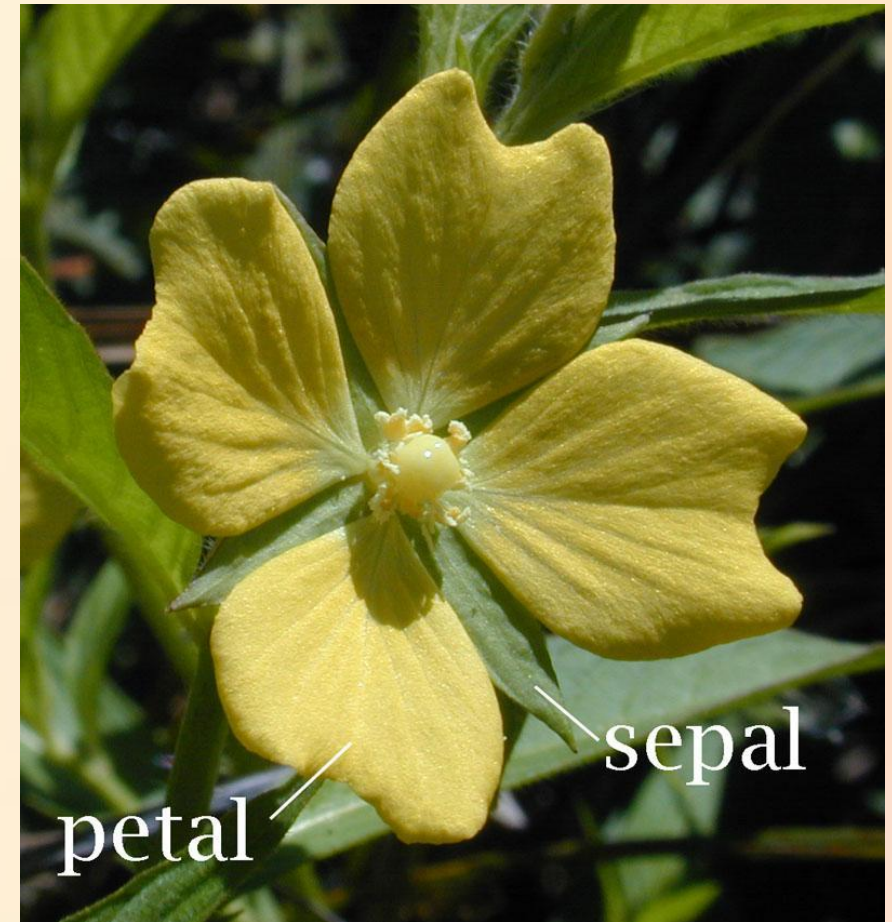


Sepals and Petals

- Often brightly colored
- These parts are sterile leaf-like structures.
 - sterile = does not have the capacity to produce gametes.

Function:

- To protect other flower parts
- To attract pollinators



Stamens

Male reproductive flower structure.

- **Anther** - pollen producing sac
- **Filament** - stalk

Function:

To produce pollen (male gamete of flower – like sperm)



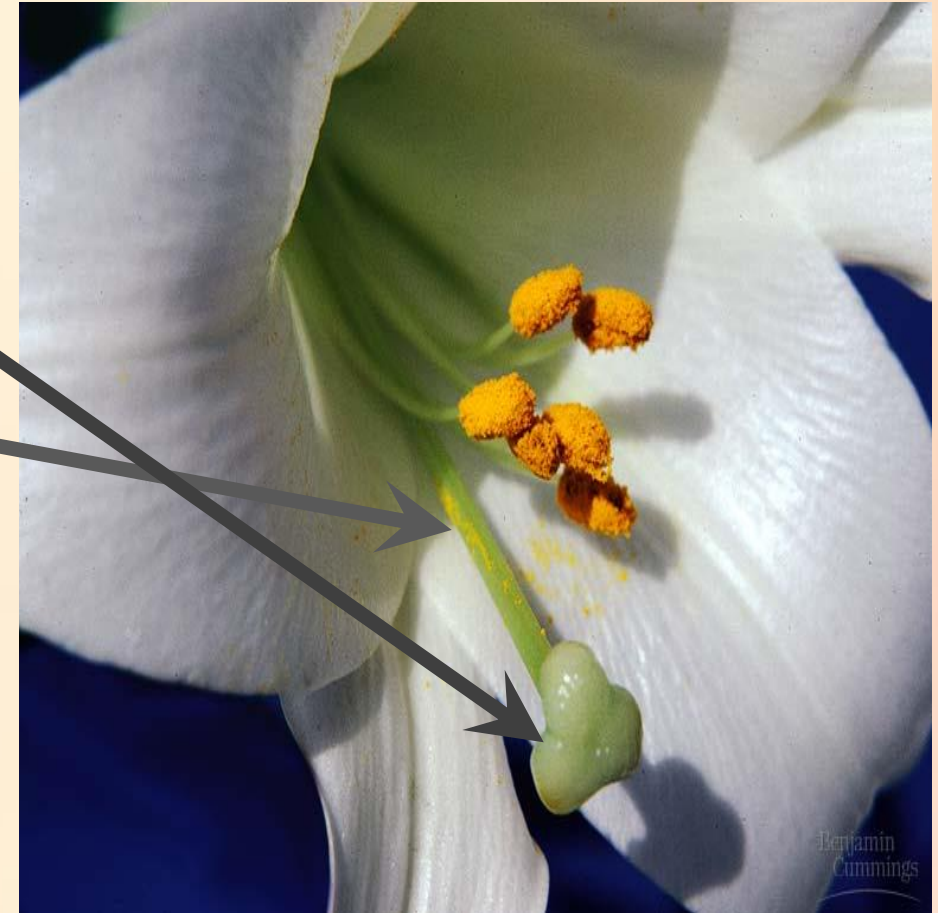
Carpel (Pistil)

Female reproductive flower structure (also known as ***pistil***).

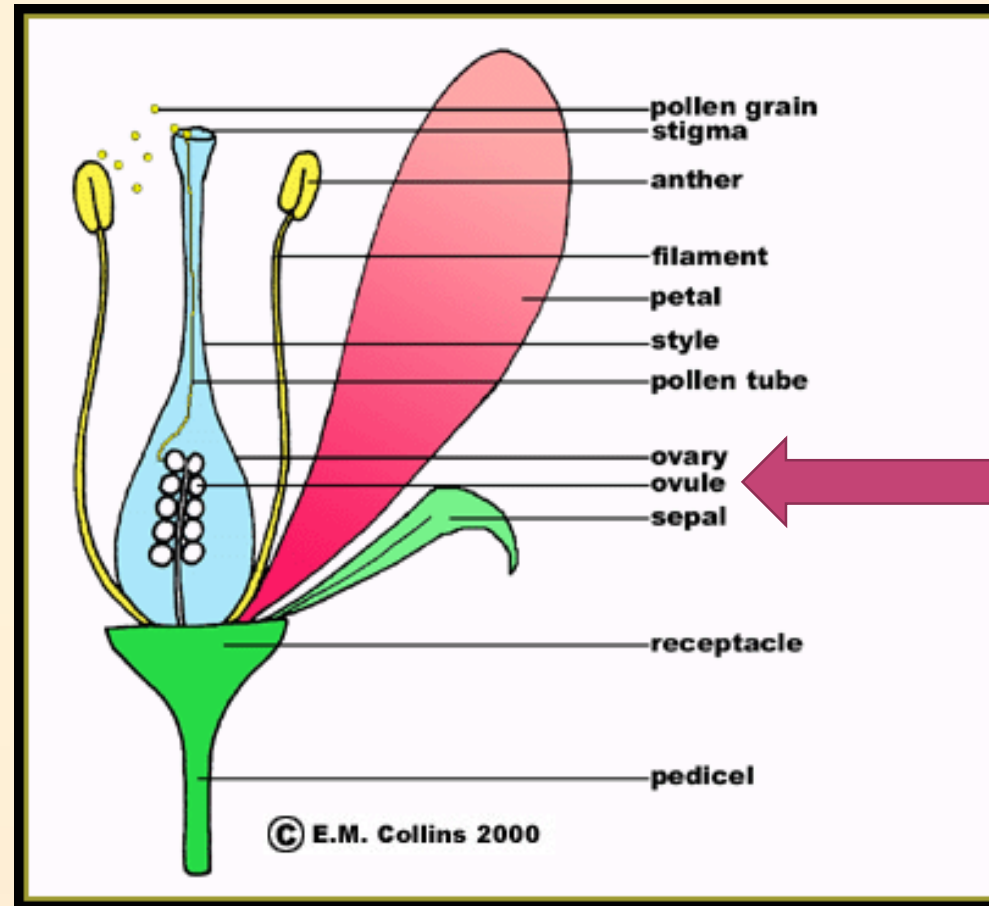
- **Stigma** – receives the pollen
- **Style** – stalk
- **Ovary** – contains the ovules

Function:

To produce ovules (female gamete of flower – like eggs)



Ovule



This is an immature seed before fertilization.

Flower Variations

- Perfect Flowers - have stamens and carpels.
- Imperfect Flowers - have stamens or carpels.



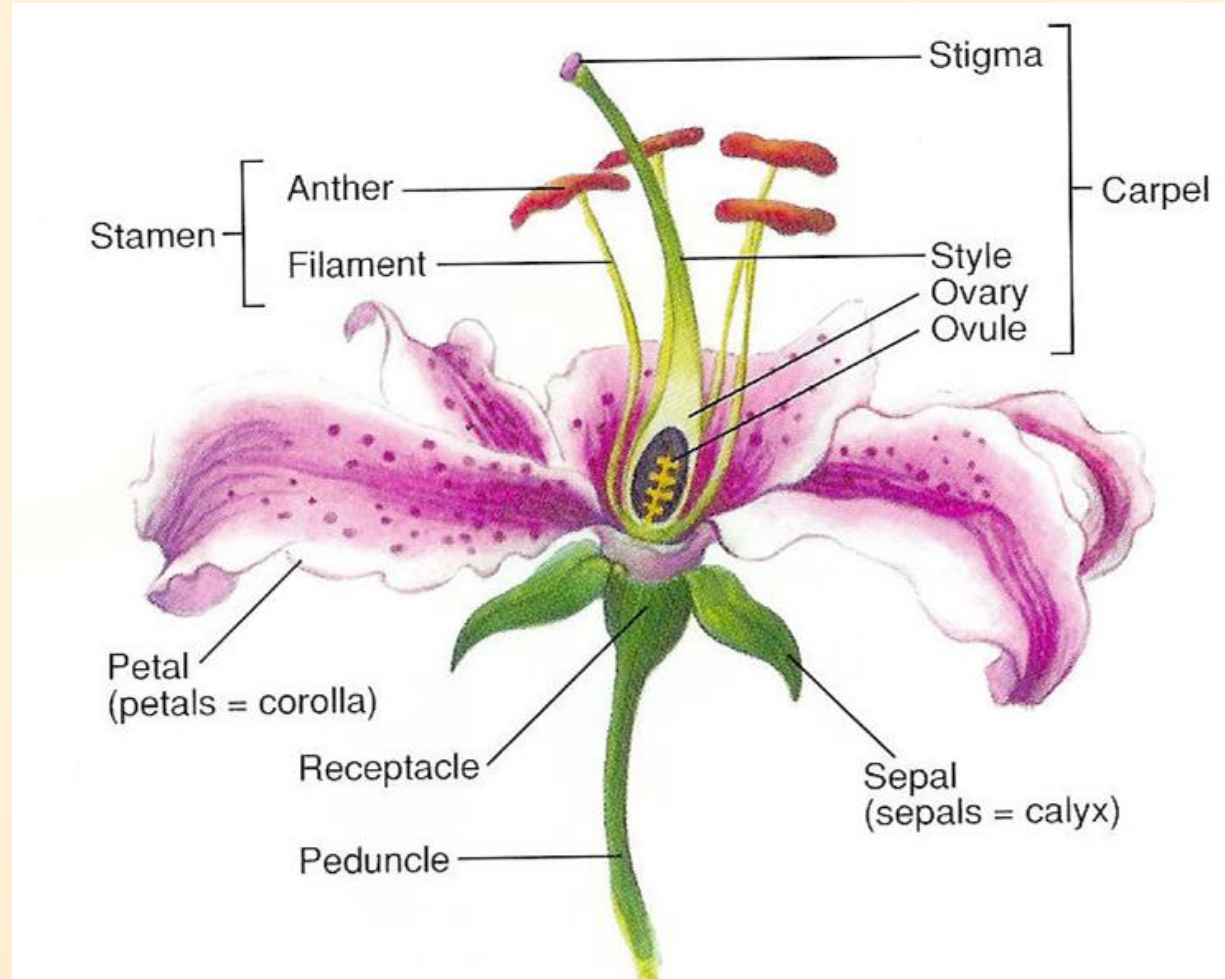
Staminate Flower



Carpellate Flower

Benjamin
Cummings

Sexual Reproduction in Plants



pollen (male) + ovule (female) → single-celled zygote → multi-celled embryo (contained in a seed) → new individual

Pollination

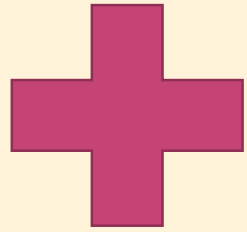
Sexual reproduction in plants is called **pollination**.



For the pollen to reach the pistil, an external source is usually required for this to occur (wind, water, animals).

Pollen Vectors

- Wind
- Water



Pollinators
(animals)

- Flies
- Moths
- Beetles
- Bees
- Butterflies
- Birds
- Bats

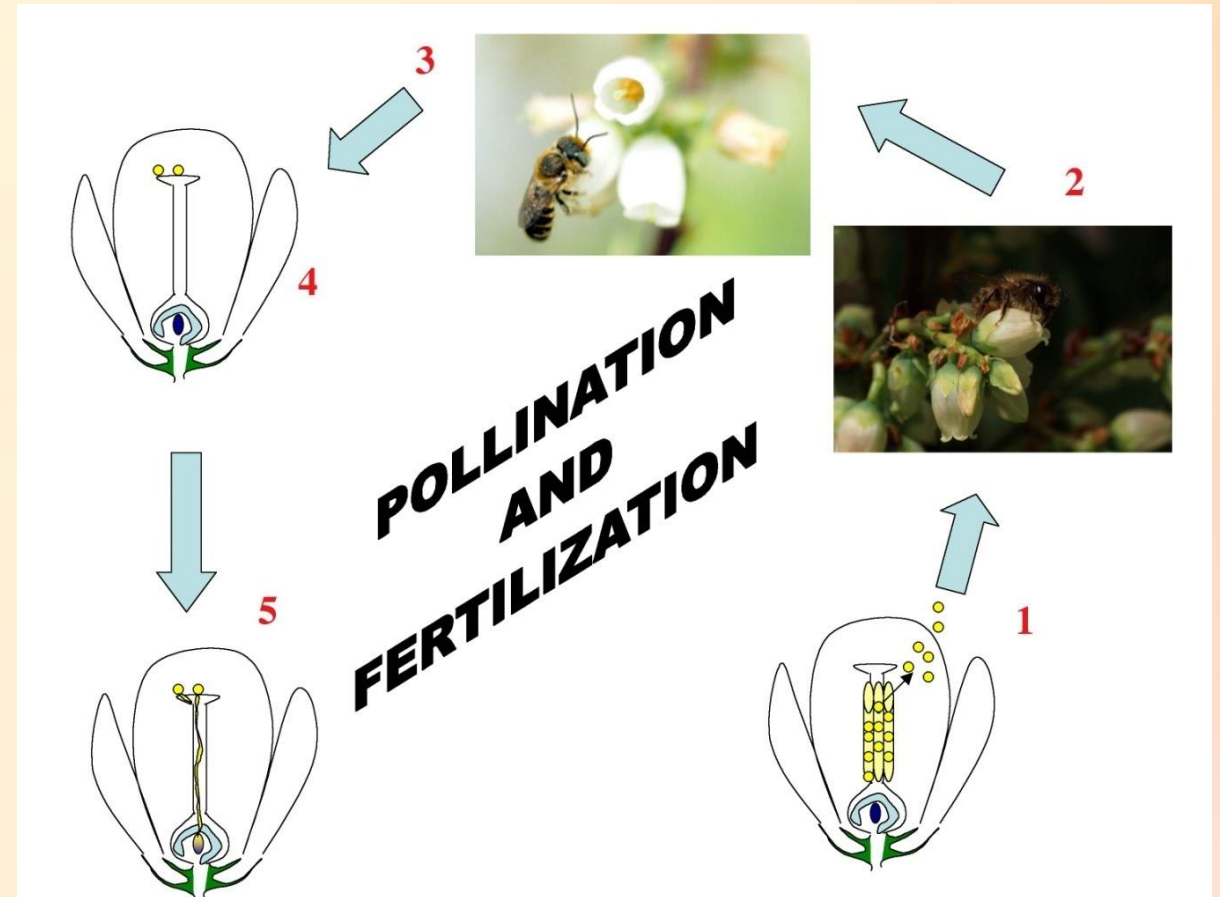


Steps for Pollination

1. The **pollen** is transferred from the **filament** to the **stigma**.

2. The **pollen** descends the **style** & reaches the **ovule**.

3. The **pollen** & **ovule** unite to produce a **seed**.



Stamen

Male reproductive organ

Anther

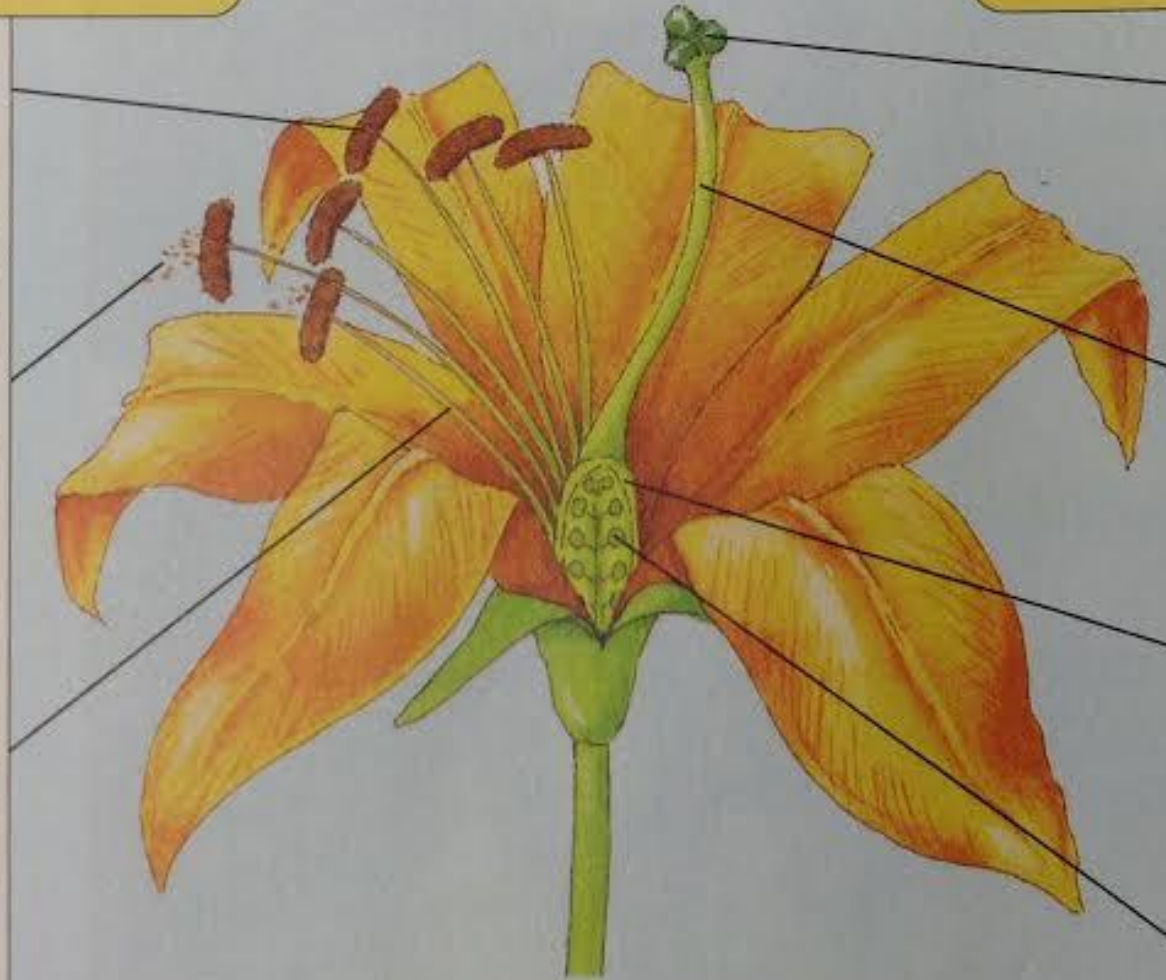
Part of the stamen where pollen is produced and stored

Pollen grains

Cases that contain male gametes (spermatozoa)

Filament

Long thin stalk that supports the anther



Pistil

Female reproductive organ

Stigma

Sticky surface of the pistil that captures pollen grains

Style

Long thin stalk that supports the stigma

Ovary

Swollen base of the pistil that contains female gametes (ovules)

Ovules

Female gametes

- Pollination leads to **fertilization**. During fertilization, the male and female gametes combine.

After the ovule has been fertilized, you end up with:

- ***Zygote*** (fertilized egg cell) → the plant embryo
- ***Ovule*** → the seed
- ***Ovary*** → the fruit

FLOWER

Ovary

Ovule

Egg cell

FRUIT

Fruit

Seed

Plant embryo

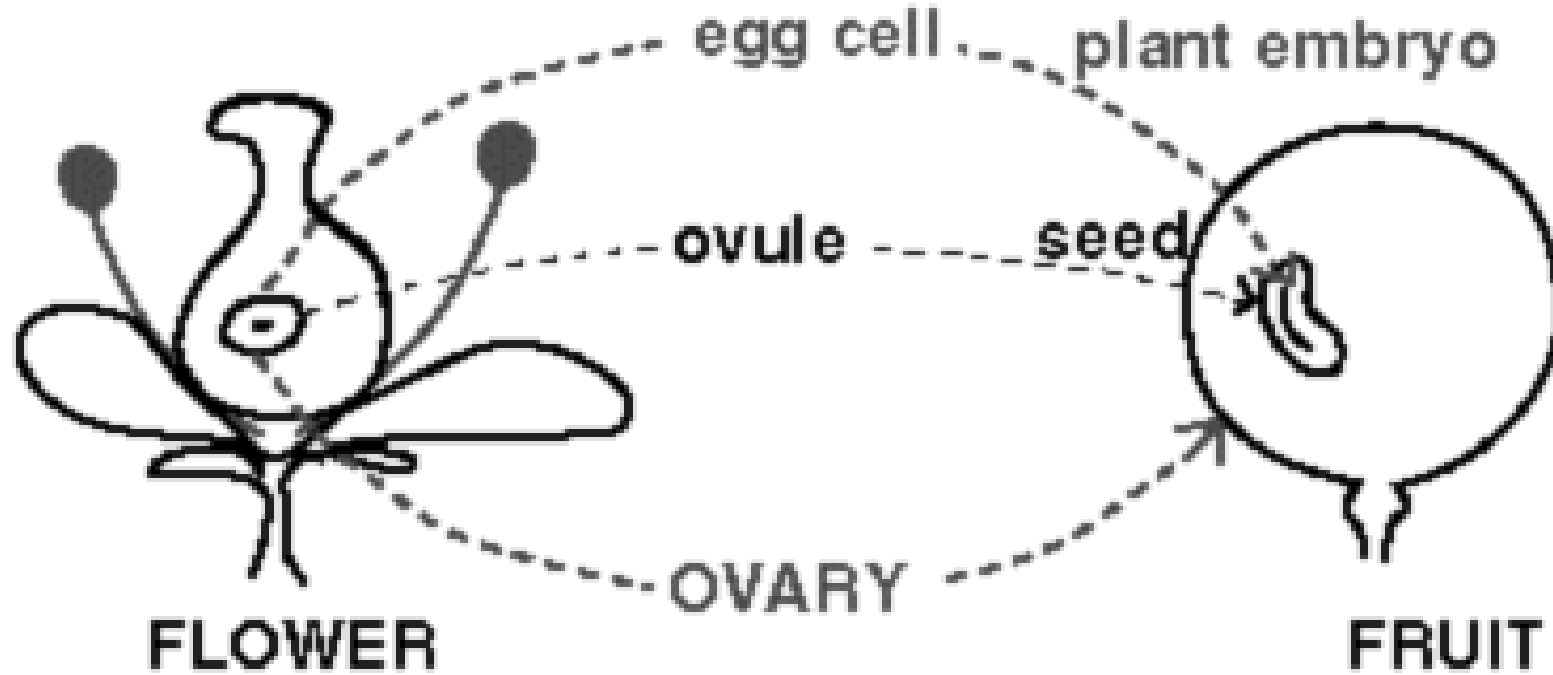


Figure 6-B: Flower/Fruit Relation

Fruits & Seeds

Fruits:

There function is to protect the seeds & to aid in seed dispersal.

Seeds:

They contain a miniature plant & are the main dispersal mechanism for plants.

Fertilization in animals

Definition: Fertilization occurs when a spermatozoon and ovum (male and female gametes) of a single species combine.

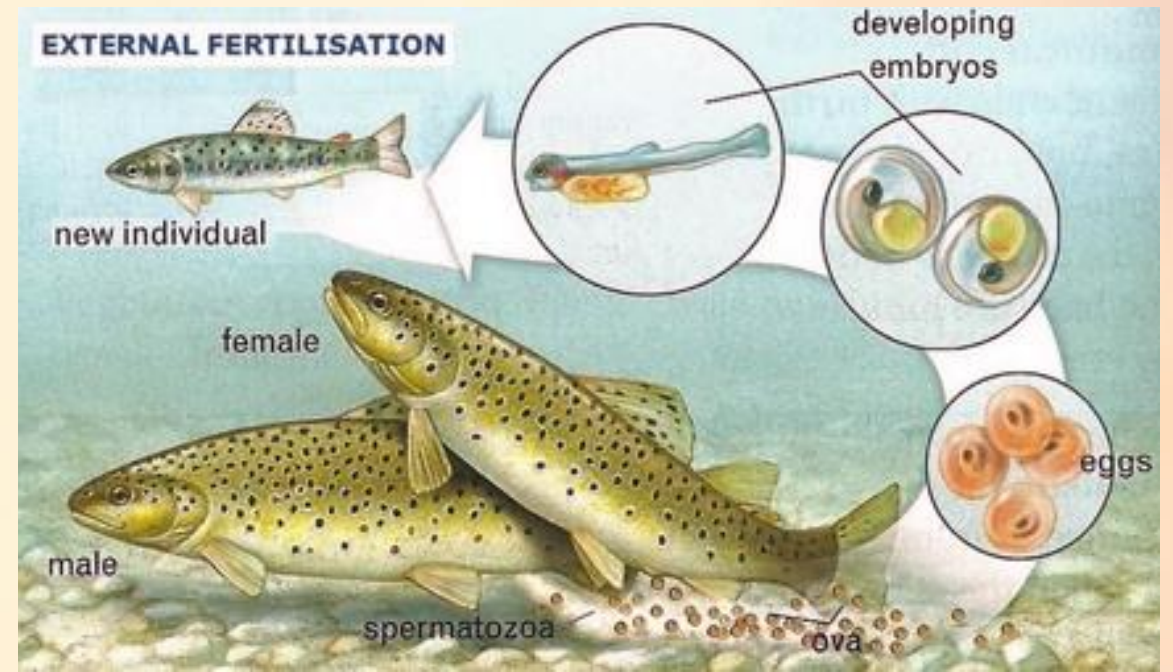
Fertilisation must take place in a **moist** environment. Why?

1. Male and female gametes are very fragile and will die if they dry out.
2. Moisture will keep the egg membrane more soft which will allow the spermatozoon to penetrate it more easily.
3. Spermatozoa can only move in a moist environment.

There are two main types of fertilization in animals:
external fertilization and **internal fertilization**.

Internal vs. External fertilization

- **External fertilization:** it is a type of fertilization where the sperm and egg unite outside of the female body. Most aquatic animals reproduce by external fertilization (frogs, fish, etc.).
- **Internal fertilization:** is it a type of fertilization where the egg and sperm unite inside the female body. Examples of animals that have this type of fertilization are mammals (humans, reptiles, whales, rabbits, dogs, etc.), and birds.



Review: Asexual & Sexual Reproduction

Recall...

Asexual reproduction:

results in offspring that are *genetically identical* to the parent organism.

- Only 1 parent

Sexual reproduction:

results in offspring that are *genetically different* from the parent organisms.

- 2 parents
- Involves gametes

Comparing the two

Asexual Reproduction

- advantages
 - does not require special cells or a lot of energy.
 - can produce offspring quickly.
 - in a stable environment, it creates a large, thriving population.
- disadvantages
 - limited ability to adapt.
 - face massive die-off events if the environment changes.

Sexual Reproduction

- advantages
 - lots of variation within a species.
 - able to live in a variety of environmental settings.
 - able to adapt to changes in the environment.
- disadvantages
 - needs time & energy.
 - produce small populations.